

Immiscible Paint-Like Liquids: 5 Approaches

Comparison

Overview: We explore five distinct two-liquid combinations that behave like **paint-thickness fluids** but remain *immiscible* (don't mix). This allows colored phases to stay separate and be reused. Each approach lists materials (with Amazon.de sources) and approximate costs, and we compare their viscosity (paint-like thickness), reusability, material availability, and cost.

Approach 1: Silicone Oil vs. PFPE (Perfluoropolyether) – “Best Technical Split”

Description: *Silicone oil* (polydimethylsiloxane) and *PFPE* fluid form an excellent immiscible pair with a **sharp separation**. PFPE liquids (e.g. **Krytox** or **Fomblin** oils) are much denser (up to ~1.8–1.9 g/mL) than silicone (~0.97 g/mL), so the silicone layer will float cleanly atop the PFPE ¹ ². They also have very low mutual solubility, meaning they won't mix even under agitation (high interfacial tension). This yields two **distinct layers** – ideal for reuse by simple decanting. To achieve **paint-like viscosity**, choose a high-viscosity silicone (e.g. 1000 cSt) and a similarly viscous PFPE (if needed, small amounts of fumed silica can thicken either phase). This approach is *laboratory-grade*: the separation performance is superb, but PFPE oils are specialized and **expensive**.

Materials (Approach 1): (with typical Amazon.de links & prices)

- **Silicone Oil (1000 cSt)** – e.g. pure silicone oil for treadmills or RC models. Price: ~€45 for 5 liters ³ or ~€20–€30 per liter.
- **PFPE Fluid (Perfluoropolyether Oil)** – e.g. *DuPont Krytox* synthetic PFPE lubricant (often sold in ~30 mL tubes). Price: ~€30 for 30 mL ⁴ (~ €1000/L, very high). Small PFPE-based spray lubes (OSiXO Krytox 30 mL) cost ~€40 ⁵.
- **Oil-Soluble Dye/Pigment (for silicone)** – *Optional*, to color the silicone phase. Oil-based dyes or pigments (e.g. cosmetic pigment) can be used. (A few euros).
- **Fluorinated Dye (for PFPE)** – *Optional*, PFPE is inert so coloring is tricky; consider using colored solid particles (or just leave one layer clear/white for contrast).

Each liquid can be colored differently (e.g. silicone tinted blue, PFPE left clear or white) for visual contrast. Both materials are **chemically inert**, so they won't react or degrade. The high cost and sourcing difficulty of PFPE make this the most “*technical*” solution, but it offers an immaculate separation (no emulsification).

Approach 2: Silicone Oil vs. Hydrocarbon Oil – “Showy & Accessible”

Description: A **silicone oil** paired with a **hydrocarbon oil** (such as mineral oil or paraffin oil) provides an easy, visually striking immiscible combination. Silicone oils are generally **immiscible with mineral oils** due to their differing polarity and molecular structure ¹. In acrylic art, these two tend to **layer rather than**

mix, producing “cells” and swirls ². This approach is **highly accessible**: both oils are common and affordable. For **paint-like thickness**, use a medium- to high-viscosity silicone (e.g. 500–1000 cSt) and a **heavy mineral oil** (sometimes called “white oil” or paraffin oil). Mineral oil is usually thinner, so you may thicken it by choosing a *viscous grade* (some mineral oils are sold as “heavy” or *medicinal grade*, slightly more viscous) or by adding a safe thickener (e.g. a little fumed silica or petroleum jelly). Color each phase with appropriate **oil-soluble dyes** for contrast (e.g. one red, one yellow).

Materials (Approach 2):

- **Silicone Oil (pure PDMS, 500–1000 cSt)** – Readily available as lubricant. *Price*: ~€20–€30 per liter ³ for ~1000 cSt pure silicone oil.
- **Mineral Oil (Paraffin/White Oil)** – Use a highly refined, clear mineral oil (lamp oil or cosmetic grade). *Price*: ~€15 per 1 L (pharma grade ~€14.95/L) ⁶. Smaller bottles (e.g. 0.5–1 L for lamps or cutting boards) are ~€5–€10.
- **Oil-Based Colorants:**
 - *For Silicone*: Any oil-soluble pigment or **silicone-compatible dye**. For instance, artists’ oil paint or oil-soluble candy colors can tint the silicone phase.
 - *For Mineral Oil*: **Oil-based food coloring** (made for chocolate/candy) works well. Sets of 10–15 colors (5–10 mL each) cost ~€9–€15 ⁷. For example, a 15-color x 6 mL oil-based dye set is ~€9.49 ⁷.

This pair is **easy to separate**: after use, let the mixture settle and you’ll see a clear layer of one floating on the other (mineral oil is usually less dense and will float on silicone if the silicone is very dense; with lighter silicones ~0.97 g/mL, mineral oil (~0.85 g/mL) might actually float *under* silicone – but either way, two layers form). You can pour off the top layer and reuse each. The **cost is low** and all ingredients are Amazon-available, making this approach very practical for demonstrations.

Approach 3: Silicone Oil vs. *Fluorosilicone* Oil – “Niche but Elegant”

Description: *Fluorosilicone oil* (a fluorinated silicone) is a specialty fluid that is **immiscible with standard silicone** oils. Fluorosilicones are essentially silicones where some methyl groups are replaced by trifluoropropyl groups, yielding a fluid with different polarity and often higher density ⁸ ⁹. This means regular PDMS silicone and fluorosilicone **won’t mix**, similar to how oil and water separate. In fact, in vitreoretinal surgery, fluorosilicone oils have been used as “heavy liquids” precisely because they stay separate from normal silicone ¹⁰. This approach is **elegant** but **niche** due to the rarity of fluorosilicone fluids outside industrial contexts. When obtained, you can pair a silicone oil (choose matching viscosity) with a fluorosilicone of similar viscosity for a nicely matched “paint-like” feel. The density of fluorosilicone (~1.3–1.4 g/mL for some types) is a bit higher than normal silicone, so typically the fluorosilicone will form the lower layer and silicone the upper.

Materials (Approach 3):

- **Silicone Oil (PDMS)** – Same as used in Approach 2 (e.g. 1000 cSt). *Price*: ~€20/L.
- **Fluorosilicone Oil** – e.g. *Dow Dowsil™ FS-1265* fluid or similar fluorinated siloxane. Often sold through chemical suppliers. *Price: Very high* – e.g. ~£311 (~€350) for 500 mL ¹¹ (≈ €700/L). Smallest quantities (100 mL) can still cost €50–€100. (Not typically on Amazon; specialized distributor required.)

- **Compatible Dyes/Pigments:** You may color these with **oil-dispersible pigments**. For example, add a pinch of inorganic pigment (iron oxide, etc.) to one phase for color. Avoid dyes that contain polar solvents (water/alcohol) as those won't dissolve in these oils.

The **viscosity** of both oils can be chosen to be similar (fluorosilicones come in grades like 300 cSt, 1000 cSt, etc. ¹²). This yields two thick, syrupy liquids that layer cleanly. **Reuse:** Excellent – they separate by density; you can reclaim each layer. **Note:** Fluorosilicone is *less accessible* and very costly – consider this only if you require a unique property (e.g. higher temperature stability or chemical resistance). Otherwise, Approach 1 or 2 can achieve similar demonstrations at lower cost.

Approach 4: Water+Glycerin Mixture vs. Mineral Oil – “Kitchen Chemistry”

Description: This approach uses a **water-based “paint”** versus an **oil-based “paint”**, both made from **food-grade ingredients**. On the water side, a mix of **water + glycerin + titanium dioxide + food coloring** creates a **thick, opaque fluid** resembling acrylic paint. Glycerin (glycerol) is a viscous liquid that increases the thickness of water (pure glycerin is ~1200 cP). By mixing in glycerin, you can tune the viscosity to a syrupy, paint-like consistency. Adding **titanium dioxide (TiO₂) powder** opacifies the mixture (making it vividly colored and opaque, like paint) and slightly boosts thickness. A few drops of **food coloring** give the desired hue. The oil phase is simply **mineral oil** (paraffin oil) colored with **oil-based candy dye** (made for chocolate). Mineral oil is immiscible with the water/glycerin mixture, so the two phases stay separate (like lava lamp fluids).

This approach is **safe, cheap, and accessible** – all components are easy to source and even edible (except you wouldn't actually eat TiO₂ in the EU). The **density difference** is significant: the water-glycerin “paint” is heavy (glycerin is dense), and mineral oil is lighter, so the oil floats on top. They can be **reused** by letting them settle and then separating the layers.

Materials (Approach 4):

- **Glycerin (Glycerol) USP** – A clear, syrupy liquid. Mix with water ~50:50 (or up to ~70% glycerin) to get a thick liquid. *Price:* ~€8–€12 per 1 liter ¹³.
- **Water** – Distilled or tap water to dilute glycerin. (Cost negligible)
- **Titanium Dioxide Powder (TiO₂)** – A white pigment (food/cosmetic grade). Add ~1–3% by volume to water-glycerin and stir to suspend (it helps mimic paint opacity). *Price:* ~€4.99 for 100 g ¹⁴ (100 g is plenty – TiO₂ is very dense, you'll use maybe a few grams).
- **Food Coloring (Water-soluble)** – Standard liquid or gel food colors for the water phase. *Price:* ~€7.59 for a set of 12×10 mL ¹⁵ (many colors). Only a few drops needed.
- **Mineral Oil** – Same as in Approach 2 (paraffin oil). *Price:* ~€7–€15 per liter (depending on grade; e.g. 1 L pharma mineral oil ~€14.95 ⁶). You can also use inexpensive **baby oil** (which is just mineral oil + fragrance).
- **Oil-Based Food Coloring** – Specialized *fat-soluble* dye for coloring oil/chocolate. *Price:* ~€9–€12 for sets of ~6–15 small bottles ⁷. (One bottle per color, but only a few drops needed.) Example: 15-color set for €9.49 ⁷.

Tips: Adjust the **thickness** by varying glycerin ratio – more glycerin yields a thicker, slower-flowing paint. A pinch of **xanthan gum** (a food thickener) could also be added to the water phase for extra viscosity, but it

may not be necessary. The mineral oil phase is naturally thinner; if you find it too runny compared to the water phase, you can slightly thicken it by dissolving a small amount of **beeswax or paraffin wax** (warm the oil to melt wax, then cool – this makes the oil more viscous). However, even without wax, the two fluids will visually behave like separate flowing paints. After use, let the mixture stand: the oil layer will collect on top (it may appear as droplets if dispersed, but will coalesce over time since no emulsifier is present), and the water-based layer sinks. They can then be poured off separately and stored for reuse. **Cost efficiency:** this is one of the cheapest approaches – mostly kitchen ingredients, roughly **€10–€20 total** for a big batch of each colored liquid.

Approach 5: Water-Based Acrylic Medium vs. Silicone Oil + Thickener – “Artist’s Recipe”

Description: This approach creates two **highly viscous, paint-like liquids** using more advanced additives, closely mimicking actual paints. The first is a **water-based acrylic medium mixture** and the second a **thickened silicone oil mixture**. They are immiscible (one water-based, one oil-based) and both can be made **very thick** (even gel-like), so they hold color well and separate cleanly.

- **Phase A (Water-based):** Start with an *acrylic gloss medium* (an acrylic polymer emulsion used in painting) as the base – this provides a glossy, viscous binder. Mix in a small amount of **xanthan gum** to further thicken the fluid to the desired consistency (xanthan readily thickens water-based liquids even at ~0.5–1% concentration). Add a concentrated **pigment** or acrylic paint to give color. The result is akin to a heavy-body acrylic paint that is still pourable.
- **Phase B (Oil-based):** Use silicone oil thickened with **fumed silica** (also known as pyrogenic silica, e.g. Evonik Aerosil). Fumed silica consists of tiny particles that strongly thicken oils by forming a network (commonly used to turn liquids into gels/grease). By dispersing a few percent of fumed silica into silicone oil (stir well), you can achieve a **paste-like or gel viscosity**. This allows the silicone phase to match the thickness of the acrylic phase. You can mix in an oil-friendly **pigment** or oil-soluble dye to color it (for example, oil paint or pigment powder). The two phases will not mix because one is a polar water-polymer mixture and the other is nonpolar silicone – they will form separate blobs or layers.

Materials (Approach 5):

- **Acrylic Gloss Medium (Fluid)** – e.g. Liquitex or Vallejo gloss medium. This is essentially acrylic resin in water. *Price:* ~€15–€20 for 500 mL (e.g. ~€18.58/0.5 L ¹⁶ or ~€14.99/L for LUKAS brand ¹⁷).
- **Xanthan Gum** – A polysaccharide thickener (food grade). Use ~1 g per 100 mL to significantly increase viscosity (sprinkle slowly and stir to avoid clumps). *Price:* ~€3.99 per 100 g ¹⁸ (very cheap; one bag will last a long time).
- **Pigment (Powder or Dispersion)** – Pure pigment powder (e.g. iron oxide, phthalocyanine blue, etc.) or concentrated acrylic pigment. *Price:* ~€2–€5 per 100 g for common pigments ¹⁹ (example: @Tec iron oxide pigment 100g ~€2.99 ¹⁹). You can also just use a few drops of high-quality acrylic *paint* or *universal tint* instead of dry pigment, since the acrylic medium will bind with it.
- **Silicone Oil** – The oil phase base. Use a medium viscosity silicone (500–1000 cSt) for easier mixing of silica. *Price:* ~€20/L (as above). You might use ~100 mL for a small batch.
- **Fumed Silica (Thickening powder)** – A very light, fluffy powder (handle gently – it can waft in air). Example: Evonik Aerosil® 200. *Price:* ~€5–€10 for ~100 g (e.g. ~€5.80 for a small jar ²⁰). 100 g is a lot

(fumed silica is low-density; 100 g ~ 5+ liters by volume). **Caution:** wear a dust mask when handling, as silica dust should not be inhaled.

- **Oil-Soluble Pigment/Dye** – As with Approach 3, color the silicone phase with a pigment that disperses in oil. Oil paints or oxide pigments can work. *(Cost negligible if you have art supplies; otherwise a few euros).*

When prepared, **each phase has a thick, paint-like consistency** and they can even hold shapes or swirls briefly before slowly flowing. On contact, they will create interesting marble effects but remain separate globules. Because both are viscous, there is minimal cross-contamination – any small amount of one phase that gets suspended in the other can be filtered out or will settle. **Reuse:** You can separate the phases by letting the mixture sit; the silicone phase (usually slightly lower density) will rise above the water-acrylic phase. The difference in density is not huge (the acrylic medium + fillers might be ~1.1 g/mL, silicone ~0.97), so give it time or centrifuge lightly if available. After separating, keep them in closed jars for next use. The materials are **fully sourceable** (art supply and chemical supply, some on Amazon Marketplace). The overall cost is moderate – roughly **€30–€50** for all components, but they yield a lot of material (multiple experiments' worth).

Comparison of Approaches

The table below summarizes key differences between the five approaches:

Approach (Fluid Pair)	Viscosity (Paint-Like)	Reuse & Separation	Availability of Materials	Approx. Cost
1. Silicone vs. PFPE <i>Lab-grade oils</i>	High – Can use very viscous oils; match viscosities as needed. Little or no thickener required.	Excellent – Very clear separation (silicone floats on PFPE) ¹ . Easy to decant and reuse.	Low for PFPE – Silicone oil is common, but PFPE fluids are specialty items (may need industrial supplier).	Very High – Silicone ~€20/L; PFPE ~€1000/L (e.g. €30 for 30 mL) ⁴ .
2. Silicone vs. Hydrocarbon <i>(Mineral oil)</i>	Moderate – Silicone oil available in various viscosities; mineral oil is thinner, but heavy grades exist. Could thicken oil slightly if needed.	Excellent – Immiscible layering (silicone and mineral oil do not mix) ¹ . Easy to separate (two clear layers).	High – Both silicone and mineral (paraffin) oils are easily bought on Amazon or in stores ⁶ .	Low – ~€20–€30 total for a liter of each oil. Dyes for coloring ~€10.

Approach (Fluid Pair)	Viscosity (Paint-Like)	Reuse & Separation	Availability of Materials	Approx. Cost
3. Silicone vs. Fluorosilicone <i>Specialty oils</i>	High – Both can be obtained in similar high viscosities (e.g. 1000 cSt). No extra thickener needed.	Excellent – Immiscible by design (fluorinated vs. non-fluorinated silicone) ⁸ . Form separate layers; reusable.	Low – Fluorosilicone oils are rare and typically only via industrial suppliers. Silicone base oil is common.	Very High – Fluorosilicone ~€300–€700/L (e.g. £311 for 500 mL) ¹¹ . Not cost-effective for casual use.
4. Water/Glycerin + TiO₂ vs. Mineral Oil <i>Food-grade ingredients</i>	Good – Glycerin-water can be made quite thick (syrupy) ¹³ . Mineral oil is thinner but small wax additives can help; usually acceptable as is.	Very Good – Strongly immiscible (like oil and water). Layers separate; glycerin phase sinks, oil floats. Easy to reuse by decanting.	Very High – All ingredients are common: pharmacy or grocery (glycerin), cooking supplies (dyes), Amazon etc.	Very Low – Perhaps €15–€25 for everything in quantity (glycerin ~€9/L ¹³ , TiO ₂ ~€5/100g ¹⁴ , oil ~€7/L, dyes ~€8).
5. Acrylic Medium + Xanthan vs. Silicone + Silica <i>Art/Chemical hybrid</i>	Excellent – Both phases can be made very thick (even gel-like) with additives. Closely mimics actual paint consistency.	Good – Immiscible (water-based vs oil). Complete separation takes more time if very viscous, but eventually layers out. Reusable after settling.	Medium – Acrylic medium and pigments from art stores (also on Amazon). Fumed silica from chemical suppliers or Amazon Marketplace ²⁰ . All obtainable, but not as household-common.	Moderate – Estimated ~€30–€50 for initial purchases, which cover many uses (medium ~€15, silica ~€5, silicone ~€10 for 0.5L, etc.).

Notes: All approaches produce fluids that **resemble paint in thickness and behavior**. Approaches 1–3 involve pairs of oils with inherently high viscosities, while 4–5 achieve thickness through additives. **Color separation** is maintained in each case because the liquids do not mix; you can swirl them together for marbling effects and then let them separate out again. The choice depends on budget and availability: for

most hobby purposes, Approaches 2, 4, or 5 are recommended due to their low cost and easy sourcing, whereas Approaches 1 and 3 are more appropriate for specialized demonstrations where the absolute clarity of separation or specific chemical properties are required (and the higher costs justified).

1 2 Does Silicone and Oil Mix Explained with Uses and Compatibility - Blissam

<https://www.blissamchem.com/does-silicone-and-oil-mix.html>

3 Silicone Oil 5 Litres 1000cst Silicone Oil - Amazon.de

<https://www.amazon.de/-/en/Silicone-Oil-Litres-1000cst/dp/B0B31XK2YV>

4 Krytox - Amazon.es

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5 Original BMW Special lubricant 30 ML KRYTOX. - HUBAUER-Shop.de

[https://www.hubauer-shop.de/en/9408523-special-lu.html?](https://www.hubauer-shop.de/en/9408523-special-lu.html?refID=HU8943622552&srsId=AfmBOopBYdmzi9bh8a0tXcHtE4AQIIs6WZoNiU5hH3gEXg4V3ljZ8VEX)

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6 Paraffinöl kaufen (Paraffinum) - Ph. Eur. - 1000 ml, 14,95 €

<https://www.vitalundfitmit100.de/paraffinum-paraffin-oel-1000ml>

7 Fettlösliche Lebensmittelfarbe 15 Farben X 6ml - Amazon.de

<https://www.amazon.de/Fettl%C3%B6sliche-Lebensmittelfarbe-Farben-6ml-Lebensmittelfarben/dp/B0DDKNJBjT>

8 9 [PDF] FL-1000 Fluorosilicone Oil - Gluespec

<https://www.gluespec.com/Materials/SpecSheet/2c98ebf8-95ec-4c4e-9435-a797f1fe8926>

10 [PDF] Factors Contributing to the Emulsification of Intraocular Silicone and ...

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11 DOWSIL® FS 1265 300cs Fluorosilicone Fluid 500ml ... - Locatory.com

<https://www.locatory.com/products/aircraft-part/consumables/FS12653004/8d98969f22db32d1176151c21060b9850791d455/c462792e-593f-4c8a-97c3-638d74e2718d?id=c8b5167e-e735-4ed5-b4d5-2b2f8cf5f2b8>

12 DOWSIL® FS-1265 Fluid - Esco Products

<https://www.escopro.com/dow-corningr-fs-1265-fluid.html>

13 Glycerine 99.5% For Making Soaps, Moisturising and ... - Amazon.de

<https://www.amazon.de/-/en/PLANTAWA-Vegetable-Glycerine-Making-Moisturising/dp/B0BJ2DDNVX>

14 Inoxia Titanium Dioxide, Weight: 100 g - Amazon.de

<https://www.amazon.de/-/en/Inoxia-Titanium-Dioxide-Weight-100/dp/B075F9NNKQ>

15 Alle Angebote / Lebensmittelfarben / Koch- & Backzutaten

https://www.amazon.de/Lebensmittelfarben-Alle-Angebote-Koch-Backzutaten/s?rh=n%3A364764031%2Cp_n_deal_type%3A26902994031

16 Acrylic medium AV: Medium fluid gloss 500 ml - Amazon.com.be

<https://www.amazon.com.be/-/en/Acrylic-medium-AV-Medium-fluid/dp/B005SBDV70>

17 LUKAS Malmittel & Co. LUKAS Farben Marken Shop - Creativ ...

<https://www.creativ-discount.de/LUKAS-Malmittel-Co.>

18 Buxtrade 100g Xanthan Gum Food Additive E 415 Stabilizer ...

<https://www.amazon.de/-/en/Buxtrade-Xanthan-Additive-Stabilizer-Emulsifier/dp/B013996RZU>

19 **tec Premium Pigmentpulver, Eisenoxid, Oxidfarbe ... - Amazon.de**

<https://www.amazon.de/tec-Premium-Pigmentpulver-Eisenoxid-Oxidfarbe/dp/B09S6Q9V7J>

20 **Aerosil 200 1 l bis 30 l Thixotropierpulver, Füllstoff, Thixotropiermi**

<https://gfkshop.de/products/aerosil-200-1l-bis-30-l-thixotropierpulver-fullstoff-thixotropiermittel-kieselsaure-pulver?srsId=AfmBOoqH6MrD1ZBvBS2n3qJRDjxT22dTPDFEISpjaRbHaDtdasPD8v0>